

How does Large Language Model Work?

- **Concept:** LLM is a prediction engine. It looks at a sequence of words and calculates which word is most likely to come next based on everything it learned during training.

The cat sat ...

The cat sat on ...

The cat sat on the ...

The cat sat on the mat

Candidate	Probability
mat	65%
rug	20%
sofa	10%
apple	0.0001%

What is Top P & Top K

- **Concept:** Before selecting the next word, an LLM ranks thousands of potential words by probability. Top-K and Top-P are cutoff filters that tell the model: "Only choose from the absolute best candidates and throw away the rest. ”
- **Top K**
 - **Behavior:** Limits the choices to a fixed number (K) of the top most likely next words.
 - **Example:** If you set Top-K = 3, the model throws away everything except the top 3.
- **Top P**
 - **Behavior:** Limits the choices based on a combined total probability threshold (P),
Example: If you set Top-P = 0.95 (95%), the model keeps adding the top highest-probability words to a list until their combined probabilities reach 90%, ”.

Temperature

- **Concept:** Temperature is a setting that controls how creative or predictable the LLM's responses will be. It ranges from 0.0 to 1.0 (or sometimes 2.0).
- **Low Temperature (0.0 – 0.3)**
 - **Behavior:** Highly accurate, conservative, repetitive, and factual.
 - **Example:** Ask low-temp AI to finish: "The sky is..." → It will almost always say "blue".
- **High Temperature (0.7 – 1.0 +)**
 - **Behavior:** Creative, surprising, diverse, but higher risk of mistakes ("hallucinations").
 - **Example:** Ask high-temp AI to finish: "The sky is..." → It might say "an ocean of twilight" or "a velvet canvas".

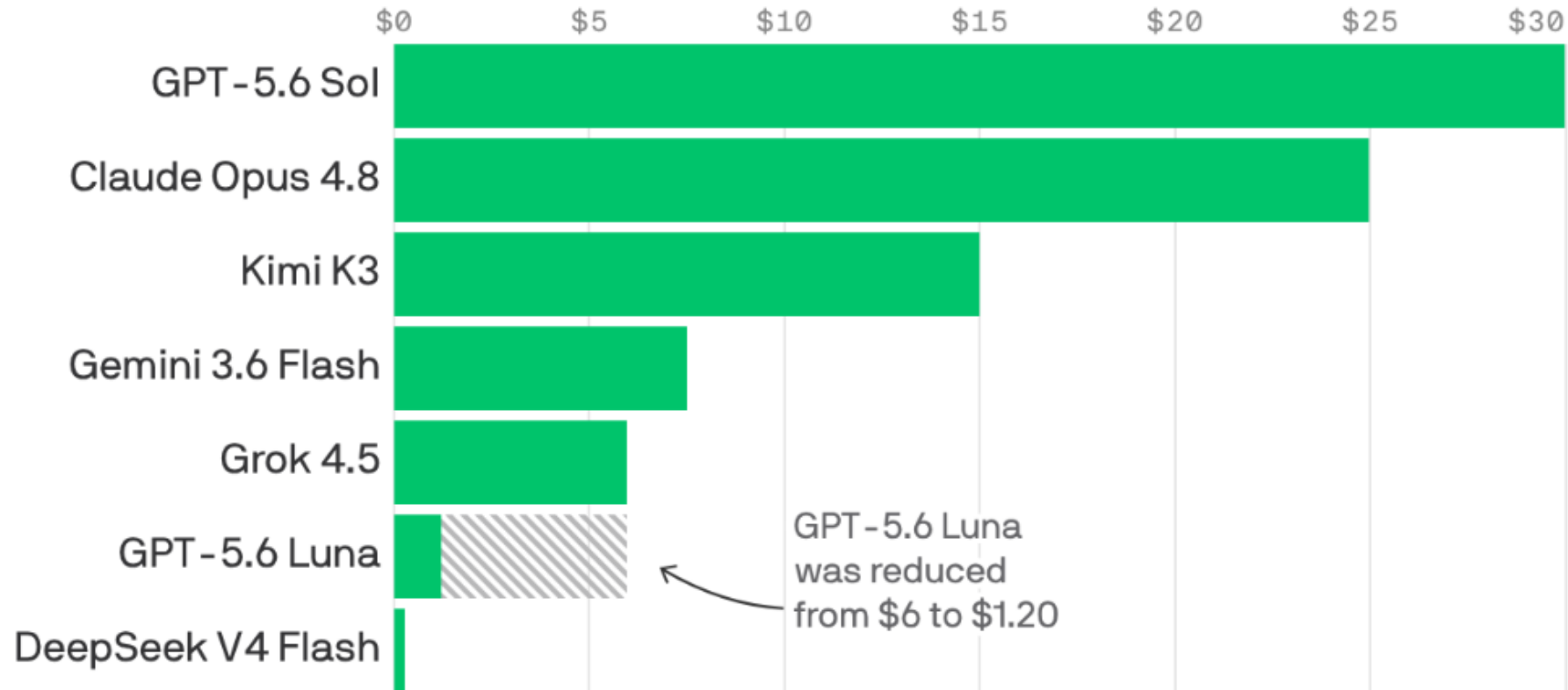
What is a Token?

- **Concept:** LLMs don't read full words the way humans do. They break text down into smaller chunks called tokens.
- **Rule:** 1 token is roughly equal to 4 characters or 0.75 words in English.
- **Examples:**
 - Short words are usually 1 token (e.g., "apple" = 1 token).
 - Long words get split into multiple tokens (e.g., "unbelievable" = "un" + "believe" + "able" = 3 tokens).
 - Punctuation, spaces, and emoji also count as tokens.

Important: *The cost of LLM's usage is charged on the basis of consumed tokens.*

Price per one million tokens of select AI model outputs

As of July 31, 2026



Data: Axios research; Chart: Sara Wise/Axios

Source: <https://www.axios.com/2026/08/01/deepseek-model-cheap-ai-price-war>

Context Window

- **Concept:** The Context Window is the maximum amount of text (measured in tokens) an LLM can **process, read, and remember** at a single time—including both your prompt and the AI's answer.
- The AI suffers from "forgetfulness." It loses track of the earliest part of the conversation to make room for new text.
- **Examples:**
 - A Window having 4,000 tokens reading a short 5-page article. The AI remembers everything on those pages, but if you give it page 6, it forgets page 1.

Context Window of Different LLMs

Provider	Latest Model Series	Context Window (Tokens)
Google	Gemini 3.5 Pro / Gemini 3.1	2,000,000 (2M)
OpenAI	GPT-5.6 Sol / GPT-5.6 Luna	1,050,000
Anthropic	Claude Sonnet 5 / Claude 3.5 Series	1,000,000 (1M)
OpenAI	GPT-4o / o1 Series	128,000 (128K)
Meta	Llama 3.3 / Llama 3.1 (Open Source)	128,000 (128K)
xAI	1. Grok 4.20 / Grok 4.1 Fast 2. Grok 4.5 Flagship	1. 2,000,000 (2M) 2. 500,000 for Grok 4.5

Model Parameters

- **Concept:** Parameters are the internal weights and connections inside an AI model that it learned during training. They represent the model's "brain size" and overall knowledge capacity.
- Expressed in billions (e.g., 7B, 70B, 175B)
- **Comparison**
 - **Small Model (7 Billion parameters):** Fast, lightweight, can run on a laptop, good for basic tasks.
 - **Large Model (70B – 175B+ parameters):** Requires powerful servers, slower, but understands complex reasoning, nuance, and logic much better.

Parameters of Different LLMs

Provider	Latest Model Series	Parameter Count
Meta	1. Llama 3.3 (70B) 2. Llama 3.1 (405B)	1. 70 Billion (open) 2. 405 Billion (open)
xAI	Grok 4 / 4.20 Series	~1.7 Trillion (MoE - Closed)
OpenAI	GPT-5 Series (GPT-5.6 / Sol / Terra)	~1.5 – 1.8 Trillion (MoE - Closed)
Google	Gemini 3.5 / 3.1 Series (Pro / Flash)	~1.2 – 1.5 Trillion (MoE - Closed)
Anthropic	Claude 3.5 / Sonnet 5 Series	~650 – 800 Billion (Closed)
xAI	Grok 4 / 4.20 Series	~1.7 Trillion (Closed)

Playgrounds

Google AI Studio:

- https://aistudio.google.com/prompts/new_chat

OpenAI Playground

- <https://platform.openai.com/chat/edit?models=gpt-5.5>

Claude Playground

- <https://platform.claude.com/dashboard>